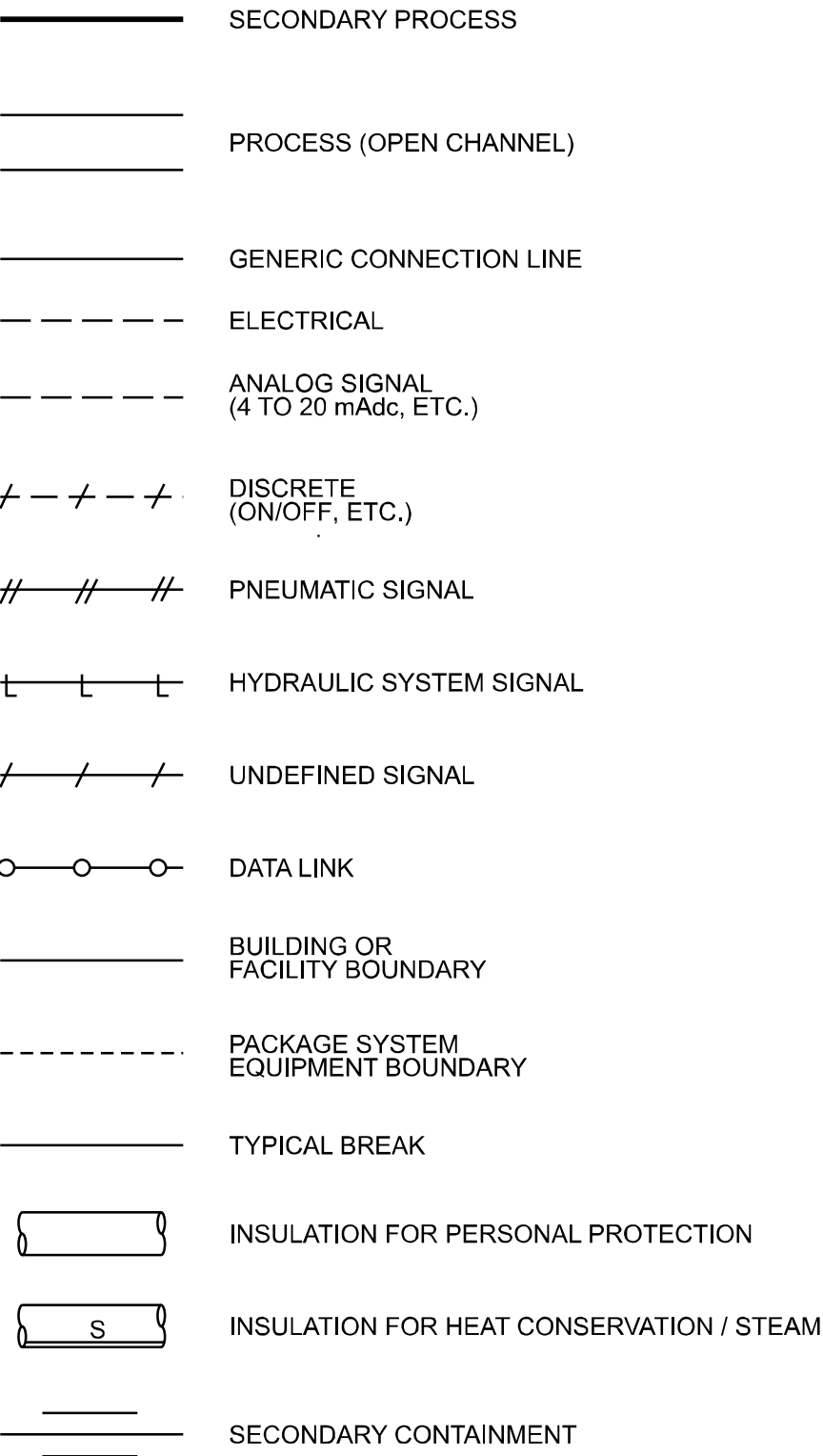


GENERAL NOTES

1. COMPONENTS AND PANELS SHOWN WITH A SINGLE ASTERISK (*) ARE TO BE PROVIDED AS PART OF A PACKAGE SYSTEM.
2. THIS IS A STANDARD LEGEND. THEREFORE, NOT ALL OF THIS INFORMATION MAY BE USED ON THE PROJECT.

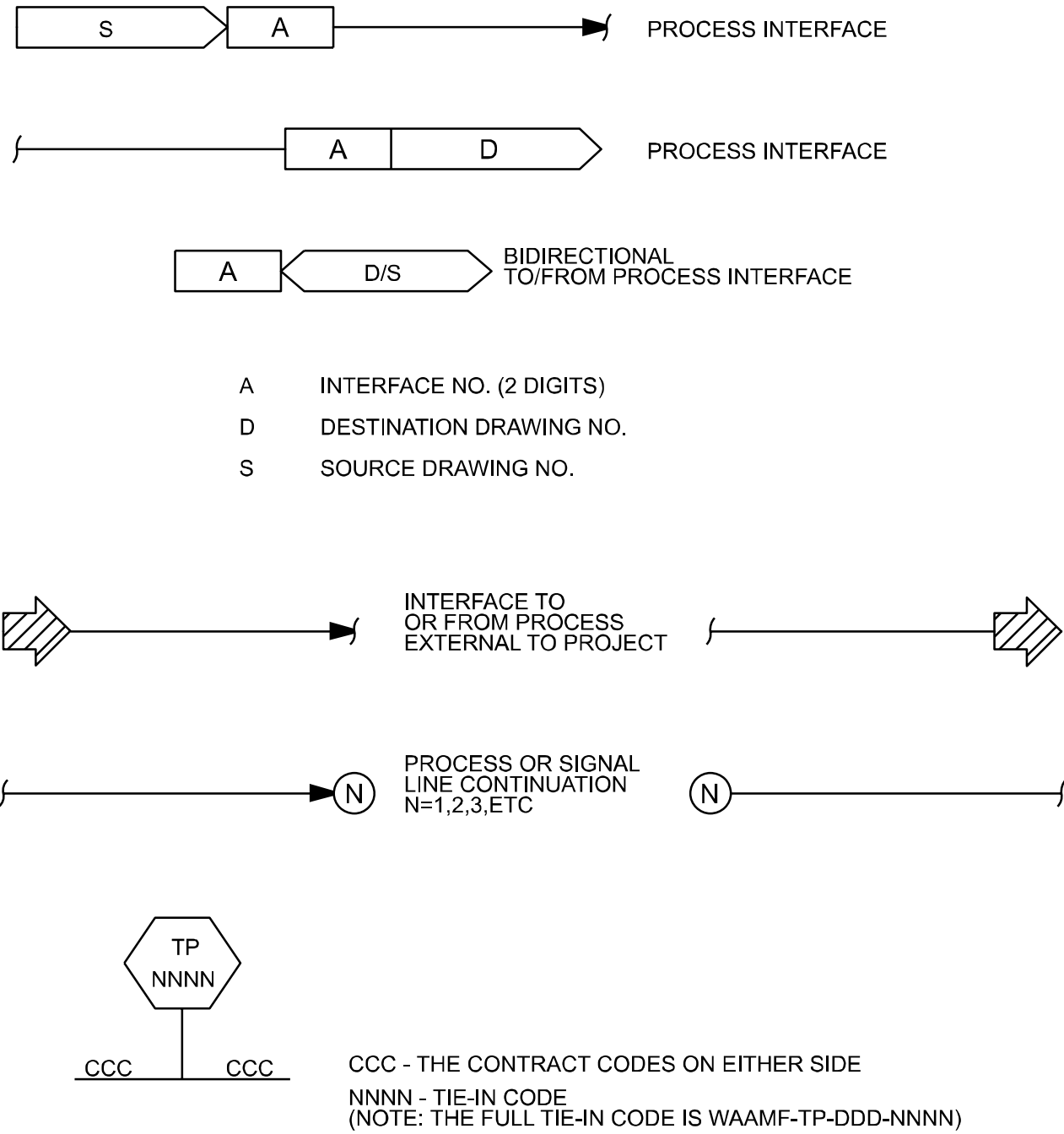
LINE LEGEND



CONNECTING LINES

NON-CONNECTING LINES

INTERFACE SYMBOLS



ABBREVIATIONS & LETTER SYMBOLS

AC	ALTERNATING CURRENT/ ACTIVATED CARBON (FOR VESSELS)
AS	ADJUSTABLE SPEED
CO	CARBON MONOXIDE
COND	CONDUCTIVITY
CS	CONSTANT SPEED
CL ₂ etc.	CHLORINE (TYPICAL: USE STANDARD CHEMICAL ELEMENT ABBREVIATIONS)
CH ₄	METHANE
COD	CHEMICAL OXYGEN DEMAND
DC	DIRECT CURRENT
DO	DISSOLVED OXYGEN
DP	DIFFERENTIAL PRESSURE
F	FLUORIDE
F.C.	FAIL CLOSED
FRC	FREE RESIDUAL CHLORINE
F.L.	FAIL LAST
FMP	FIELD / MCC / PLC
FNI	FIELD NETWORK INTERFACE
F.O.	FAIL OPENED
FOS	FAST-OFF-SLOW
FOSA	FAST-OFF-SLOW-AUTO
FOSR	FAST-OFF-SLOW-REMOTE
FP	FIELD / PLC
FR	FORWARD-REVERSE
H	HIGH
HARD	HARDNESS
HH	HIGH-HIGH
HOA	HAND-OFF-AUTO
HOR	HAND-OFF-REMOTE
H ₂ S	HYDROGEN SULPHIDE
ISB	INTRINSIC SAFETY BARRIER
L	LOW
LEL	LOWER EXPLOSIVE LIMIT (FLAMMABLE GAS DETECTION)
LC	LOCKED CLOSED
LO	LOCKED OPEN
LL	LOW-LOW
LOS	LOCKOUT STOP
LR	LOCAL-REMOTE
MA	MANUAL-AUTO
MC	MODULATE-CLOSE
MCC	MOTOR CONTROL CENTER
MLSS	MIXED LIQUOR SUSPENDED SOLIDS
MP	MCC / PLC
MSC	MANUFACTURER SUPPLIED CABLE
NC	NORMALLY CLOSED
NH ₂ CL	MONOCHLORAMINE
NH ₃	AMMONIA
NO	NORMALLY OPEN
NOX	NITRATE/ NITRITE
OC	OPEN-CLOSE(D)
OCA	OPEN-CLOSE-AUTO
OCR	OPEN-CLOSE-REMOTE
OO	ON-OFF
OOA	ON-OFF-AUTO
OOR	ON-OFF-REMOTE
ORP	OXIDATION REDUCTION POTENTIAL
OSC	OPEN-STOP-CLOSE
P	PHOSPHATE
PART	PARTICLE COUNTER
PD	POSITIVE DISPLACEMENT
pH	HYDROGEN ION CONCENTRATION
PLC	PROGRAMMABLE LOGIC CONTROLLER
RIO	REMOTE I/O UNIT
RTU-X	REMOTE TELEMETRY UNIT NO. X
SBD	SLUDGE BLANKET DETECTOR
SDI	SILT DENSITY INDEX
SF	SLOWER-FASTER
SS	START-STOP
TRC	TOTAL RESIDUAL CHLORINE
TOC	TOTAL ORGANIC CARBON
TOD	TOTAL OXYGEN DEMAND
TOX	TOXICITY
TS	TOTAL SOLIDS
TSS	TOTAL SUSPENDED SOLIDS
TURB	TURBIDITY
VHC	VOLATILE HYDROCARBONS
VIB	VIBRATION
VOC	VOLATILE ORGANIC COMPOUND
Δ	DIFFERENCE
Σ	SUM
x	MULTIPLY
÷	DIVIDE
F(X)	CHARACTERIZED
x ⁿ	RAISED TO THE Nth POWER
√	SQUARE ROOT
AVG	AVERAGE
1:1	REPEAT OR BOOST
>	SELECT HIGHEST SIGNAL
<	SELECT LOWEST SIGNAL
}	BIAS
%	GAIN OR ATTENUATE
TP	TOTAL PHOSPHORUS

TENDER

T1 20/05/22	ISSUED FOR TENDER C4B	CN	GOT
T1 02/07/21	ISSUED FOR TENDER C3D2	CN	GOT
T1 18/12/20	ISSUED FOR TENDER C3B1	CN	GOT
T1 23/07/20	ISSUED FOR TENDER C2B	CN	GOT
T1 24/04/20	ISSUED FOR TENDER C3A	CN	GOT
T1 08/04/20	ISSUED FOR TENDER C3D1	CN	GOT
REV	DATE	DESCRIPTION	REVIEWED /AUTHORISED

CLIENT:



SINGAPORE'S
NATIONAL
WATER AGENCY

PROGRAMME MANAGER:



AECOM

A JOINT VENTURE DELIVERING THE DTSS2 PROJECT

CONSULTANT:



CH2M HILL Singapore Pte Ltd

PROJECT TITLE:

TUAS WATER RECLAMATION PLANT (TWRP)
DTSS PHASE 2

DRAWING TITLE

P&ID LEGEND
GENERAL

DRAFTED: MW	DESIGNED: CSY	SHEET SIZE: A1
DRAFTING CHECK: DW	DESIGN CHECK: JL	SCALE: NTS
DATE: 29/11/17	REVIEWED: CN	PBS: 131101
CONTRACT:	DRAWING NUMBER: PUB-DTSS2-TWRP	
-	SHEET NUMBER: WSW00-P-0001	REVISION: T1

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